

Aaron Wu

Software Engineer

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EDUCATION

University of California, San Diego

M.S. Computer Science, GPA: 3.77/4

San Diego, CA

Expected Grad March 2026

University of California, Santa Cruz

B.S. Computer Science, GPA: 3.90/4

Santa Cruz, CA

Sept 2020 – June 2024

EXPERIENCE

Lawrence Livermore National Labs

Software Engineering Intern

Livermore, CA

June 2025 – Sept 2025

- Developed search features for a data repository application using Angular, Spring Boot, Solr, and Postgres, improving data discoverability for users
- Extended one-box search to support multiple ID fields by defining new Solr schema fields, enhancing metadata search service, and wiring new Angular UI components
- Implemented permission facets by extending indexing services to include access-control metadata, allowing users to filter search results by access level
- Added metadata-scoped prefix matching to the one-box search, allowing users to find items by partially typed metadata values

jLab in Smart Sensing, UCSC

Full Stack Developer

Santa Cruz, CA

Mar 2023 – Sept 2024

- Built a data monitoring platform using React, Flask, and Postgres to ingest and analyze data from a sensor network
- Engineered a REST API with Flask to log voltage, current, and temperature sensor data into a Postgres database
- Constructed an interactive data dashboard with live graphs using React, featuring filter and zoom capabilities
- Developed support for simultaneous large CSV file exports using Redis and Celery for distributed task processing
- Implemented a CI/CD pipeline with GitHub Actions and Docker to automate testing and deploy APIs to AWS

Tech4Good, UCSC

Undergraduate Research Fellow / Software Engineer

Santa Cruz, CA

June 2022 – June 2024

- Led development of a web application for collaborative, AI-assisted interview transcript annotation using Angular, NgRx, RxJS, and Firebase, adopted by 100+ students each quarter in a business strategy course
- Built end to end UI components and data services to deliver real time crowdsourced AI feedback for students
- Developed an admin dashboard for use by class instructors to manage homework assignments and project teams
- Improved Firestore database schema structure to optimize page loads, reducing database reads by 40%

PROJECTS

Memory Profiler | TypeScript, Rust

github.com/aaron-wu1/Pulse

- Built a local-first macOS memory profiler in Rust and TypeScript for real-time process analytics
- Implemented a Rust backend to parse kernel virtual memory statistics and parameters for memory usage data
- Constructed a TypeScript frontend to display real-time process and memory statistics in a sortable table
- Integrated embedded pgvector and Ollama to deliver local RAG-based process explanations

jsSQL | JavaScript

github.com/aaron-wu1/jsSQL

- Created a npm library for a file-based SQL database engine supporting SQLite queries
- Added support for query types including selects, joins, aggregations, and filters
- Implemented B+ tree indexing to speed up query performance

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, Rust, Go, C, SQL, HTML, CSS

Backend: Spring Boot, Flask, Redis, Postgres, Solr, Firebase

Frontend: React, Angular, Next.js, RxJS, Redux

Infrastructure & Tools: Docker, Kubernetes, Amazon Web Services (ECS, EC2, CloudFront), CI/CD (GitHub Actions)